

Wangjie Su

Undergraduate | College of Engineering | Peking University

+86 188 6803 2449 | wsu0605@stu.pku.edu.cn

<https://github.com/csxd0605> | <https://csxd0605.github.io/>

Research Interests

Current focus: Reliable AI agents for scientific simulation and industrial workflows, with an emphasis on domain-tool and knowledge integration, long-horizon memory and context, multi-agent orchestration, failure recovery, and evaluation.

Future directions: Grounding long-horizon ReAct agents in multimodal perception, sensor streams, world models, and physical action to build perception–planning–action–feedback loops for autonomous laboratories, scientific instruments, and general-purpose robots.

Education

Peking University, B.S. in Theoretical and Applied Mechanics

Sep 2023 – Present

Peking University, Double Degree in Intelligent Science and Technology

Sep 2024 – Present

- **GPA:** 3.818/4.0
- **Selected Coursework:** Data Structures and Algorithms (**100**), Programming in AI (**100**), Computer Vision, Probability and Statistics, Numerical Analysis, Generative Modeling, Intelligent Robotics, Trustworthy ML, LLMs and NLG, and Physics Simulation for Computer Graphics.

Research Experience

AI + Fusion Energy Series Project

Undergraduate Researcher

Mar 2026 – Present

Xiniao Technology Innovation Center, Peking University | *PIs:* Prof. Yunfeng Liang and Prof. Songfang Huang

- **Problem:** Developing AI agents that can carry out long-horizon fusion-energy simulation workflows, from task planning and knowledge retrieval to tool execution, result interpretation, and iterative refinement.
- **Implementation:** Using **Nexgent** as the experimental harness to connect domain knowledge and simulation tools through MCP, coordinate specialist agents and explicit workflows, and preserve sessions, checkpoints, permissions, and tool traces.
- **Research focus:** Investigating long-context retrieval and recovery, sustained adherence and progress in long-horizon tasks, and permission-gated failure recovery so scientific runs remain state-consistent, auditable, and reproducible.

Combustion Laboratory, Peking University

Undergraduate Researcher

May 2025 – Present

PI: Prof. Zheng Chen

- **Background:** Studying fundamental combustion problems in energy conversion and propulsion, including flame dynamics, chemical kinetics, transport, ignition, flame propagation and stabilization, and laminar flame speed.
- **Current work:** Exploring mathematical formulations of multiscale combustion processes from conservation laws and reaction mechanisms, together with machine-learning approaches to predicting key quantities and approximating complex processes.
- **Methodological focus:** Comparing the regimes of mechanistic and data-driven models, with attention to physical consistency, interpretability, data efficiency, and out-of-distribution generalization toward physics–data integrated combustion modeling.

Internship Experience

Hangzhou Yuanxi Technology Co., Ltd.

AI Agent Developer Intern

Jul 2026 – Present

Product: **ManySelves** — An Independently Designed and Built General-Purpose AI Agent Task Runtime

- **Product ownership:** Independently designed and implemented **ManySelves** end to end from scratch as a stable runtime for interchangeable document-defined agent teams; identity documents specify roles, boundaries, I/O, and handoffs, while Skill documents encode reusable methods and quality criteria.
- **Core capabilities:** Built a unified Main Agent surface with specialist routing, document context, streaming execution, multi-provider integration, typed state, checkpoints, rollback, and resumable decisions; tools and workflows connect project files, reviews, and final artifacts.

- **Service interface:** Wrapped the complete Agent runtime with FastAPI and exposed only Codex Work-like conversation and task-submission capabilities; model calls, long-horizon workflows, state transitions, and artifact management remain server-side, keeping clients simple and isolated from internal tools.

Selected Projects

Nexgent: Scientific Agent Harness *Independent Project* **May 2026 – Present**

- **Problem:** Long-running scientific tasks need more than a chat interface: agents must connect to domain tools, coordinate specialized work, survive interruptions, expose risky actions, and leave a traceable experimental record. Nexgent was developed as the Agent Harness for prototyping these requirements in the AI + Fusion Energy project.
- **System:** Built a model-agnostic, frontend-neutral runtime shared by a PyQt6 desktop workspace, Textual TUI, and headless CLI; integrated 38 tools, automatic MCP tool bridging, multi-model routing, and sub-agent/workflow orchestration in pipeline, parallel, and phased modes.
- **Workflow functions:** Implemented progressive context compression, typed project/reference memory, session save–resume–fork, checkpoint rollback, permission gates, lifecycle hooks, goals, and background tasks for controlled and recoverable experiment execution.

oh-my-opencode (now oh-my-openagent) *Open-Source Contributor* **May 2026**

- **Design rationale:** Introduced a Core–Adapter boundary and downward-only dependency DAG to separate reusable policy from OpenCode Hook, provider, and UI side effects, allowing Codex, Pi, and Claude Code adapters to share the same core behavior.
- **Novel abstractions:** Consolidated agent-name mapping, ordering, and compatibility into zero-dependency agent-config-core; modeled fallback as a per-session idle/armed/exhausted state machine with injected reachability and logging, keeping policy pure, testable, and harness-agnostic.
- **Compatible implementation:** Preserved original import paths through re-export shims and handled cleanup, post-success re-arming, chain exhaustion, and per-session notification deduplication; after maintainer discussion, the extraction design was adopted upstream and integrated into the new monorepo packages architecture.

Where Is My Codex (Flow Rescue) *Course Project* **Spring 2026**

- **Physics:** Co-developed a 2D fluid-puzzle game and implemented Position-Based Fluids in Python/Taichi with uniform-grid neighbor search, density constraints, terrain collisions, and GPU/CPU backends.
- **Engineering:** Built five playable levels and a Tiled import pipeline; added headless tests and profiling for digging, collision, channel stability, goal collection, grid construction, and constraint solving.

Tiny PyTorch *Course Project* **Fall 2024**

- **Framework:** Built a PyTorch-style tensor and reverse-topological autodiff engine with core operators, fully connected layers, convolution, pooling, softmax loss, and end-to-end MLP/CNN training.
- **Acceleration:** Implemented C++/CUDA storage, transfers, custom kernels, and cuBLAS-backed computation with Python bindings; trained on MNIST and compared single-GPU and DataParallel workflows.

Honors & Awards

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- National Scholarship and Merit Student, Peking University (2025)
 - Second Prize, Beijing Division, National College Student Mathematical Modeling Competition (2025)
 - Outstanding Learning Award and Third Scholarship, Peking University (2024)
 - First Prize, Beijing College Student Mathematics Competition (2024)

Skills

Programming: Python, C/C++, CUDA, MATLAB, LaTeX, CMake, Taichi

Tools & Frameworks: PyTorch, PyQt6, Git, Linux, Scikit-learn, Jupyter, OpenFOAM

Languages: Chinese (native); English (CET-6: 613)